Is anyone interested in being a grader next quarter??
Contact me! ☺

CSE 150: Introduction to Computer Networks

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Link Layer

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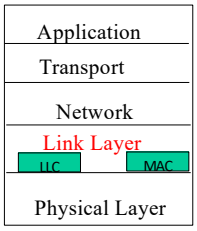
Chapter 6 &7: Link layer, LANs, Switches and WiFi

Reading as needed to support lecture material…

- 6.1 introduction, services
- 6.3 multiple access protocols
- 6.4 Switched LANs
  - 6.4.1 addressing, ARP
  - 6.4.2 Ethernet
  - 6.4.3 Link-layer switches
- 7.1 Introduction
- 7.2 skim first page only
- 7.3 WiFi: 802.11 Wireless LANs (focus on class discussion)



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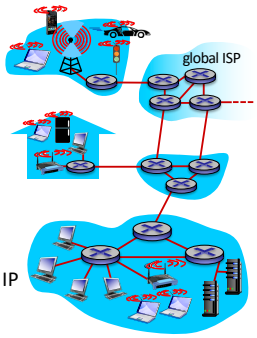
Link layer: introduction

Terminology:

- Links: Communication channels that connect adjacent nodes along communication path:
  - wired links
  - wireless links
  - LANs considered a single link in terms of forwarding (can use MAC addresses)
  - MTU for each link type on path
- Frame: Layer-2 packet:, encapsulates IP datagram



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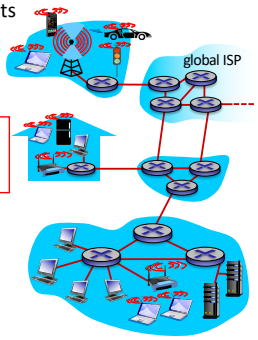
Link layer: introduction

- Protocol: Each type of link may have its own protocol!
- Ethernet and WiFi

Data-link layer has responsibility of transferring datagram from one node to *physically adjacent* node over a link



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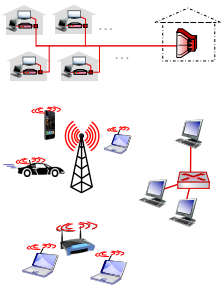
Link layer Protocol: Framing and Link Access

- Framing, link access:
  - encapsulate datagram into link layer frame, adding header, trailer
  - "MAC" addresses in frame headers identify source, destination (different from IP address!)
  - channel access if shared medium



Link Layer: 6-9

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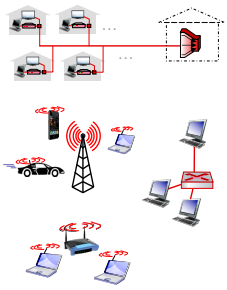
Link layer: services (Jobs)

- reliable delivery between adjacent nodes
  - we already know how to do this!
  - seldom used on low bit-error links
  - wireless links: high error rates
    - *Q:* why both link-level and end-end reliability?
- flow control:
  - pacing between adjacent sending and receiving nodes
- error detection:
  - errors caused by signal attenuation, noise.
  - receiver detects errors, signals retransmission, or drops frame
- error correction:
  - receiver identifies *and corrects* bit error(s) without retransmission
- half-duplex and full-duplex:
  - with half duplex, nodes at both ends of link can transmit, but not at same time



Link Layer: 6-10

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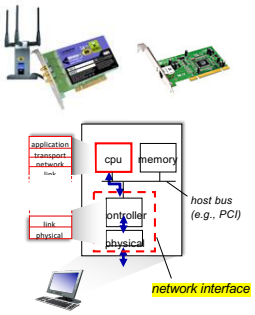
NIC: Lots going on here!

Network Interface Card (NIC) /adaptor/controller in every host and router

- Link layer implemented in NIC
  - Ethernet, WiFi
- Attaches into host's system bus
- Combination of hardware, software, firmware
- Jobs:
  - Framing
  - Link protocol
  - Link Access
  - Error Detection



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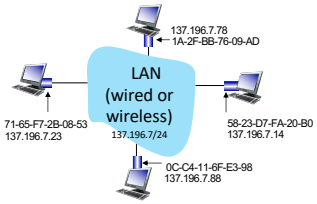
MAC Addresses. e.g.: 1A-2F-BB-76-09-AD

hexadecimal (base 16) notation (each "numeral" represents 4 bits)

adapter

Each interface has:

- unique 48-bit MAC address
- unique 32-bit IP address
- port number



- Use MAC address used "locally" to get frame from one interface to another physically-connected interface → No router!

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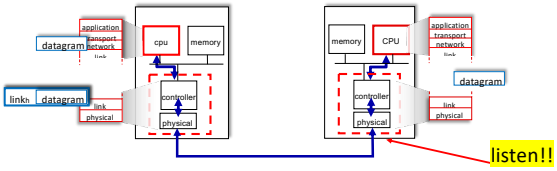
Interfaces communicating



Sending side:

- encapsulate datagram in link layer frame
  - →MAC Address
- adds error checking bits, reliable data transfer, flow control, etc.
- Sends out onto link

Receiving side:

- Checks frames for MAC address match
- If accepted, look for errors, reliable data transfer, flow control, etc.
- extracts datagram, pass to upper layer at receiving side

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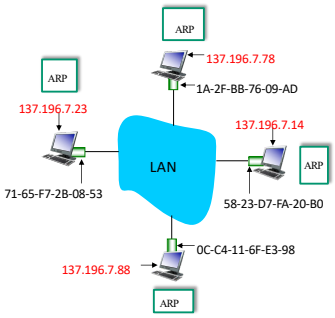
ARP: Address Resolution Protocol

Fundamental translation IP → MAC

Question: How to determine a host's MAC address? (must know destination IP address)



ARP Table: each IP node (host, router) keeps an ARP table:

IP	MAC	TTL

TTL (Time To Live): mapping will be forgotten (typically 20 min)

- WHY?

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ARP in action

- Same Network (LAN/subnet) only!!!
- How is this different from DNS?

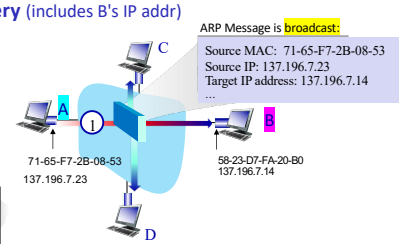
Example: A wants to send datagram to B

A's ARP table does not have B's MAC Address:

① A broadcasts **ARP Query** (includes B's IP addr)

ARP Message is broadcast:

Source MAC: 71-65-F7-2B-08-53
Source IP: 137.196.7.23
Target IP address: 137.196.7.14
...



ARP table in A

IP	MAC	TTL

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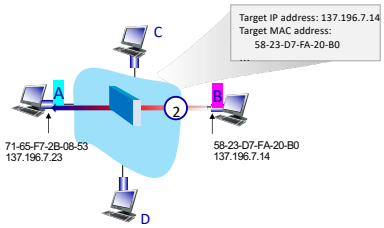
ARP protocol in action

② B replies directly to A in ARP message, giving its MAC address

Target IP address: 137.196.7.14
Target MAC address:
58-23-D7-FA-20-B0
...



③ After receiving reply, A enters B's info into its ARP table

ARP table in A

IP addr	MAC addr	TTL
137.196.7.14	58-23-D7-FA-20-B0	500

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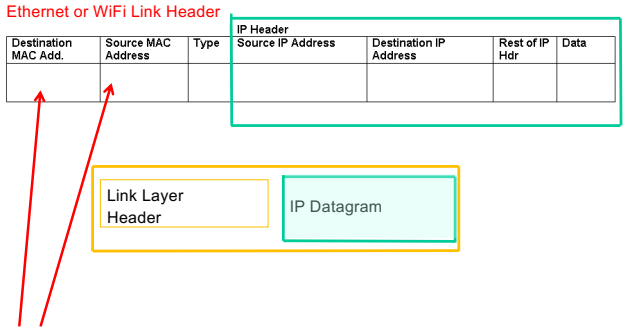
Link Layer Frame Encapsulates IP Payload



Used to reach destination on same link
Every link layer frame has its own header – you can look up the format!

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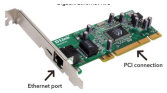
Network Interface Card

AKA Adaptor/Interface

Forwarding packets over a link

- NIC to NIC!!



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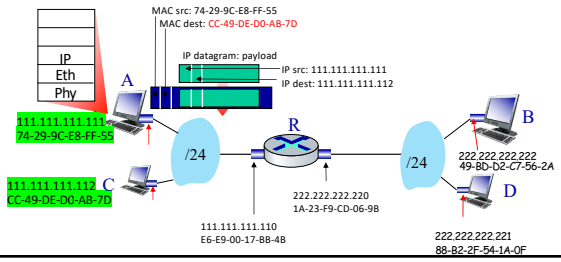
Forwarding packets on same subnet A→C

Walkthrough Example Hosts on the same LAN are reachable by MAC address

- ?? Does source know: a)destination on same LAN? B) its MAC Address

Encapsulation:

- Network Layer → IP datagram: IP Addresses
- Link Layer → encapsulates datagram & creates LL frame: MAC Addresses
- ?? Is the router involved??



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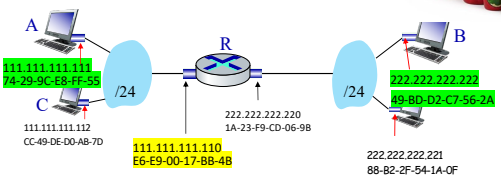
Forwarding to the universe!

Walkthrough: Sending a datagram from *A* to *B*

- Focus on addressing – IP (datagram) and Link layer (frame)
- Assume that:
  - A knows B's IP address
  - A knows IP address of first hop router, R (how?)
  - A obtains R's MAC address (how?)
- Over how many links will IP datagram travel?
- Link layer header replaced each hop!
  - Target the NIC!



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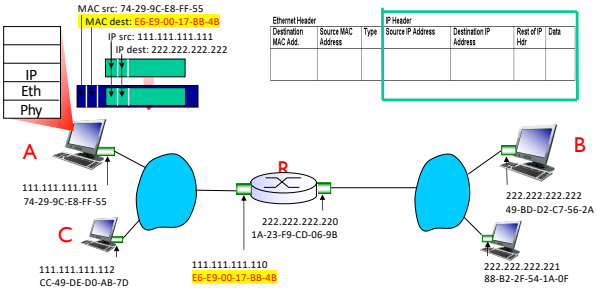
Multihop Forwarding A→B: enters Link 1

- Host A creates IP datagram: IP Addresses: src A, dest B
- NIC in Host A creates link-layer frame with Router's MAC address and encapsulates IP datagram as payload
- NIC to NIC



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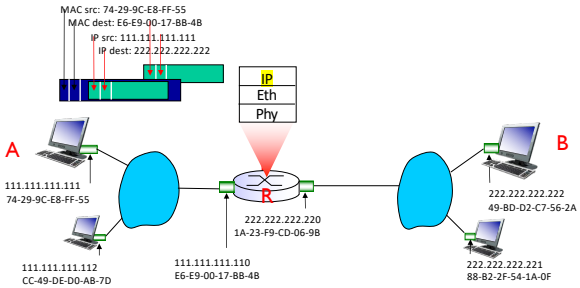
Multihop Forwarding A→B: enters Router

- Router accepts LL frame
- Decapsulates LL frame → IP datagram passed up to IP
- What's next?



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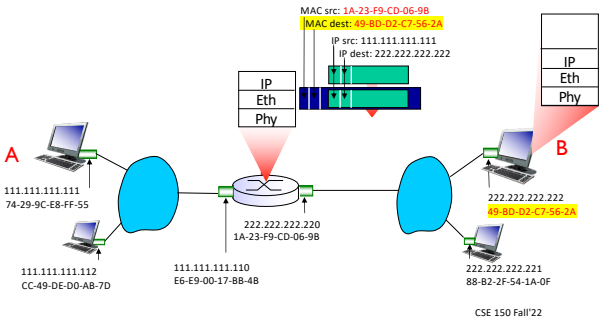
Multihop Forwarding A→B: exits Router

- IP Forwarding: Determine outgoing link
- Create NEW Link Layer Header
  - Dest- B's MAC address; Source - R's MAC address
  - IP payload? → unchanged!

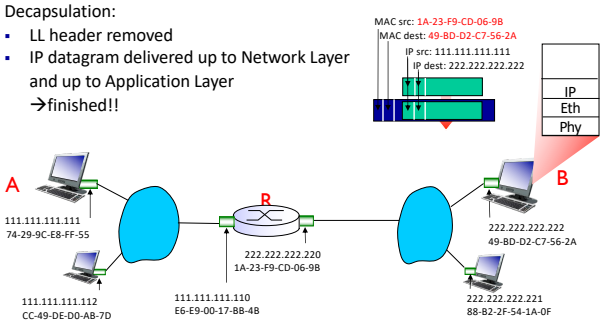


Multihop Forwarding A→B: Link 2 and arrival at Destination

- LL frame arrives at B
- MAC address matches B's NIC

Decapsulation:

- LL header removed
- IP datagram delivered up to Network Layer and up to Application Layer →finished!!



Some Big Questions about the Link Layer:

- ❒ ✓How do we send packets from one host to another over a physically connected link?
- ❒ ✓ Is the packet destination on the local LAN or outside?
- ❒ ✓ How to find MAC address

1. What are some different Link Layer protocols to support different link types?
2. How are links shared? i.e. more than one user wants to transmit over same link
3. What is the difference between a Switch and a Router?

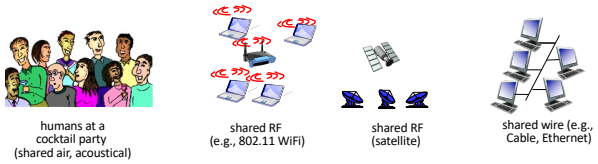
Two Types of Link Connections

Point-to-point

- PPP for dial-up access
- Between Ethernet switch and host or between routers
- Full Duplex

Single shared link

- IEEE 802.3 Ethernet
  - HFC (Hybrid Fiber Coaxial AKA Cable) and switch
- IEEE 802.11 Wireless LAN



Shared link (wired)

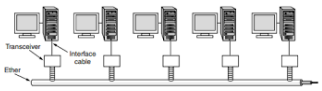


Figure 4-13. Architecture of classic Ethernet.

Shared link (wireless)

Multiple Access Control (MAC)

Multiple users: whose **turn** is it to transmit?

- Shared Links - need a plan for access
- Collision: Two or more simultaneous transmissions

MAC protocol:

- Defines how nodes share a single channel
- Communication about channel sharing uses channel itself!
  - no out-of-band channel for coordination

MAC protocols: 3 broad classes

Channel partitioning

- Divide channel into smaller "pieces" (time slots, frequency)
- Allocate piece to node for exclusive use

"Taking turns"

- Nodes take turns, but nodes with more to send can take longer turns

Random access

- Stations follow rules to "seize" channel
- no entity is in charge
  - No *a priori* coordination among nodes
- "recover" from collisions

Well known Random Access protocols

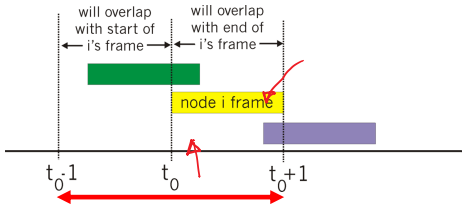
- Aloha Protocol
- Slotted Aloha
- CSMA
  - CSMA-CD → Ethernet (IEEE 802.3)
  - CSMA-CA→ WiFi (IEEE 802.11)

- The question: How to gain access to the channel, detect and recover from "collisions"?

Pure ALOHA

P(success by given node) =0.18

- 1970s University of Hawaii, low cost radio network
- Simple, no synchronization
- When frame arrives simply transmit immediately
- Collisions happen over an interval – "vulnerability period":
  - A frame transmitted at t_0 collides with any frame sent during $[t_0-1,t_0+1]$
  - Entire frame lost



Assume frame transmission is one unit of time

Pure ALOHA efficiency

Efficiency: successful transmissions over time, large N

P(success by given node) = P(node transmits) ·

P(no other node transmits in $[t_0-1,t_0]$ ·

P(no other node transmits in $[t_0,t_0+1]$ ·

$= p \cdot (1-p)^{N-1} \cdot (1-p)^{N-1}$

$= p \cdot (1-p)^{2(N-1)}$ find optimal p

… choosing optimal p and then letting N $\rightarrow \infty$

$= 1/(2e) = .18$ Really low!

How good is the protocol? Efficiency

The long-run **fraction** of time during which frames are being transmitted on the channel without collisions, *i.e.*, "successful"

Assume:

- large number of active nodes
- each node has a large number of frames to send

The problem with Aloha

Large vulnerability period

- Efficiency is limited by frequent collisions
- packet radio network → cannot listen for collisions
- Entire frame transmitted and lost

Slotted ALOHA

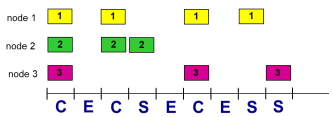
Assumptions:

- All frames same size
- Time divided into equal size slots (time to transmit 1 frame)
- Nodes start to transmit only at beginning of slot
  Nodes are synchronized
- If 2 or more nodes transmit in slot, collision

Operation:

- When node obtains new frame, transmits in next slot
  - *if no collision:* node can send new frame in next slot
  - *if collision:* node retransmits frame in each subsequent slot with prob p until success

randomization – *why?*



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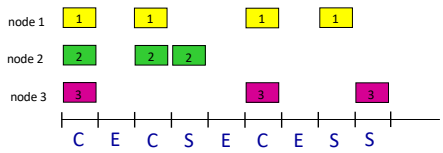
Slotted ALOHA

P(success by given node) =0.37



C: collision
S: success
E: empty

Pros:

- Simple
- Highly decentralized: only slots in nodes need to be in sync
- Single active node can continuously transmit at full rate of channel
- Improved efficiency

Cons:

- Wasted slots
  - Collisions
  - Idle slots
- Clock synchronization

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Slotted ALOHA: efficiency

Suppose: N nodes with many frames to send, each transmits in slot with probability *p*

- prob that given node has success in a slot = $p(1-p)^{N-1}$
- prob that *any* node has a success = $Np(1-p)^{N-1}$
- Max efficiency: find p^* that maximizes $Np(1-p)^{N-1}$
- For many nodes, take limit of $Np^*(1-p^*)^{N-1}$ as *N* goes to infinity, gives:

max efficiency = 1/e = .37

at best: channel used for useful transmissions 37% of time!

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CSMA (Carrier Sense Multiple Access)

An improvement for wired links!

Based on politeness!!

Human analogy: don't interrupt others! Speak only during silent periods

Listen before transmit:

- if channel sensed idle: transmit entire frame
- if channel sensed busy: defer transmission

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Carrier Sense Multiple Access (CSMA)

- The efficiency of the ALOHA protocols is limited by the large vulnerability period of a packet
  - inability to listen for collisions → packet radio network
- Carrier Sensing: Listen before transmitting
  - Stations try to reduce the vulnerability period. This is the basis of CSMA (Kleinrock and Tobagi, UCLA, 1975)
- Later Bob Metcalfe, inspired by Aloha, expands the work and introduces CSMA-CD (Ethernet)

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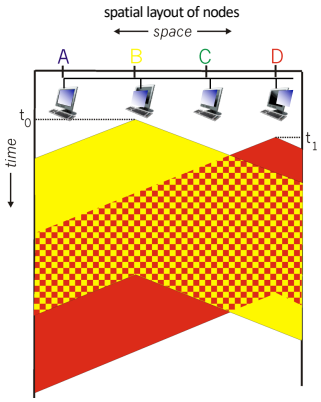
CSMA still has collisions

Collisions *can* still occur:
Nodes don't hear each other's transmission instantaneously!

- Why?

Collision: entire packet transmission time wasted

- distance & propagation delay play role in collision probability



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Improving CSMA:

- Later Bob Metcalfe, inspired by Aloha, expands the work and introduces CSMA-CD
  →Ethernet Protocol
  IEEE 802.3
- Key idea: Listen while transmitting also!

CSMA/CD (Collision Detection)

Carrier Sensing, Multiple Access → listen before transmitting

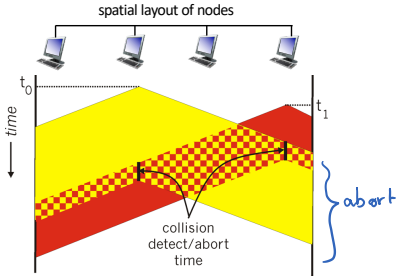
- Ethernet's MAC protocol
- Collisions *detected* within short time
- Colliding transmissions aborted, reducing channel waste

Collision Detection → listen while transmitting

- Easy in wired LANs: measure signal strengths, compare transmitted, received signals
- Difficult in wireless LANs: received signal strength overwhelmed by local transmission strength

Human analogy: the extremely polite conversationalist

CSMA/CD (reduces collision interval)



- Transmission aborted as soon as collision detected
- Channel will be idle (available) sooner

CSMA/CD has 3 States

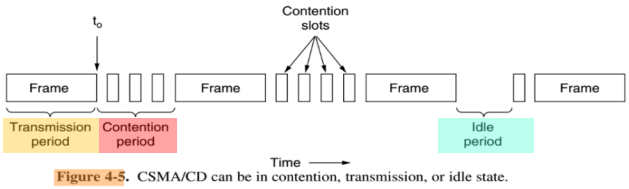


Figure 4-5. CSMA/CD can be in contention, transmission, or idle state.

Three states:

1. Successful Transmission (no collisions)
2. Idle (channel free, no stations are transmitting)
3. Contention (there has been a collision, stations back-off, spread out retransmissions)

CSMA/CD in action:

1. The NIC receives datagram from network layer, creates frame
2. NIC listens to channel:
   - senses channel idle?
     - start frame transmission.
   - senses channel busy?
     - Defer, continue sensing
3. If NIC detects no collision over entire transmitted frame, NIC is done!
4. If a collision is detected, NIC **aborts** and sends jam signal
5. NIC runs *binary exponential backoff* to decide when to try again

- after *m*th collision, NIC chooses *K* at random from *{0,1,2, …, 2^m-1} and* waits K·512 bit times ("slots")
  - returns to Step 2

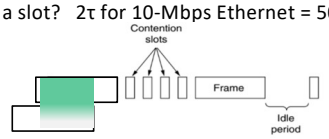
- Backoff interval increases with more collisions

After a Collision: Binary Exponential Back-off

(wait a random # of slots before re-transmitting)

- 1st collision: wait 0 or **1** slots
- 2nd collision: wait 0, 1, 2, or **3** slots (chance of collision decreases)
- 3rd collision: wait 0, 1, 2, 3, 4, 5, 6, or **7** slots
- 4th collision: wait 0, 1, 2, 3, . . ., or **15** slots
- nth collision: wait 0, 1, 2, 3, . . ., or **2^n** – 1 slots

- How long is a slot? 2τ for 10-Mbps Ethernet = 50-µsec



Random Access Protocols:

- Aloha – just go for it: 18% efficiency
- Slotted Aloha – slot boundaries 36%
- CSMA – listen before transmitting
- CSMA-CD -- listen before and
- during transmission

Increasing efficiency

Study:

- Why does CSMA still have collisions?
- Why is each protocol an improvement?

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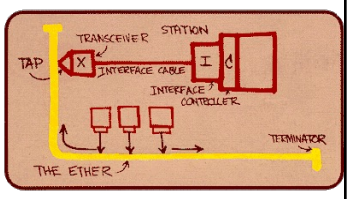
Ethernet

"dominant" wired LAN technology, rooted in Aloha, but improved

- cheap $20 for NIC
- first widely used LAN technology → FAST
- simpler, cheaper than token LANs and ATM
- 100m link (twisted pair copper), can extend
- MAC protocol: *CSMA/CD*
- Collisions handled by *Binary Exponential Backoff*



Bob Metcalfe's Ethernet sketch

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Ethernet Frame Structure (link layer)

- Sending adapter (NIC) encapsulates IP datagram in Ethernet frame
- *6 byte Source and Destination MAC Addresses*
  - NIC accepts frames with matching destination MAC address, or with broadcast address (e.g. ARP packet)
    - De-capsulate and pass datagram to Network Layer
  - Otherwise, NIC discards frame
- *Type:* Higher Layer Protocol (0x0800 IPv4; 0x0806 ARP)
- *CRC:* cyclic redundancy check at receiver
  - Error detected: frame is dropped

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Too many collisions…

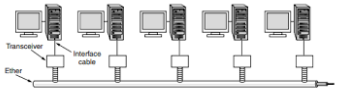

Figure 4-13. Architecture of classic Ethernet.

The Ethernet Switch

Collision Free!

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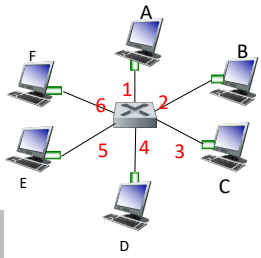
Ethernet Switch: Self-Learning →MAC Address

- *Numbered Ports*
- *Collision-free Switching:* Multiple simultaneous transmissions through switch
- *Switch is Transparent*
  - Hosts are unaware, just plug in
- *Plug-and-play, Self-learning*
  - No configuration needed
  - This is really cool!

How does switch learn Destinations?

- Observe incoming frame's SOURCE MAC address
- Build Switch Table
- Selectively forward frame based on table entries



switch with six interfaces (1,2,3,4,5,6)

CSMA/CD?

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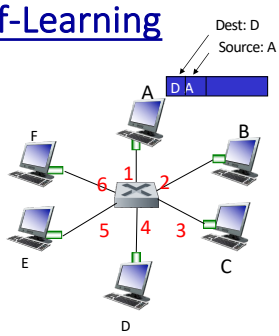
Switch: Source Self-Learning

Q: How does Switch learn?

- The facts:
  - A is reachable via port 1
  - D is reachable via port 4
- When frame arrives at switch, it remembers incoming port of the **Sender (source)**
- Records sender/port pair in Switch Table



Example: A → D

Switch table (initially empty)

MAC addr	port	TTL

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Switch: Learning/Forwarding

For each incoming data frame:

1. Learning: Observe incoming frame's Source MAC address and port number → enter into Switch Table

2. Forwarding:

```
if Destination MAC Address is in Switch Table{
    if destination port matches incoming port
        then drop frame  //optimization – already seen
    else forward frame over port indicated in Table
}
else flood  //over all interfaces
```

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Ethernet Switch Example

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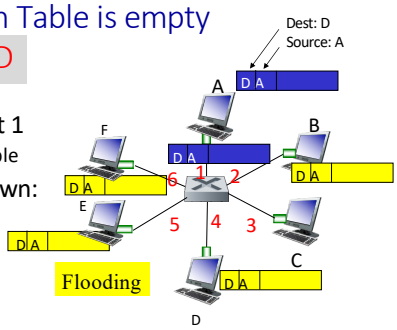
At startup, Switch Table is empty

1st transfer: A → D

1. Frame arrives at port 1
   - Enter A in Switch Table
2. Destination D unknown: →FLOOD



- Remember:
  - MAC addresses, not IP
  - Does switch know if frame is delivered?

Switch Table (initially empty)

MAC addr	port	TTL
empty		
A	1	60

Learns →

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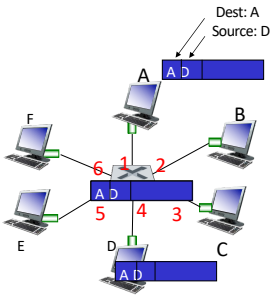
Example:

2nd transfer: D → A

1. Frame arrives at port 4
   - Enter D in Switch Table
2. Check Table: A is known!
   →Send over port 1 only

No Flooding



Switch Table (initially empty)

MAC addr	port	TTL
A	1	60
D	4	60

Learns →

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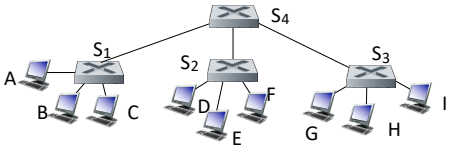
Interconnecting switches

Switches can be connected together

- Transparent, don't know about each other, but learn …



Q: A → G - how does S_1 forward frame to G?

A: Flooding and self learning! (works exactly the same as in single-switch case!)

- Now G → A …

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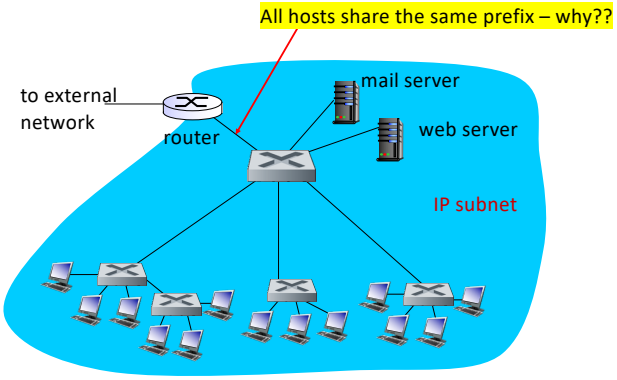
Institutional network



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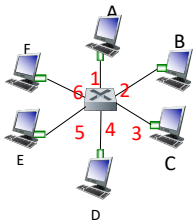
Practice!!

Action	Entries in Switch Table	Link(s) over which frame is forwarded	Explanation
B sends a frame to E	Switch learns…		
E replies with a frame to B			
A sends a frame to B			
B replies with a frame to A C broadcasts			



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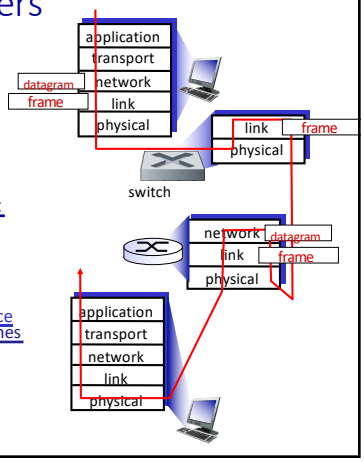
Switches vs. routers

Both are store-and-forward:

- *Routers:* network-layer devices (examine network-layer headers)
- *Switches:* link-layer devices (examine link-layer headers)

Both have Forwarding Tables:

- *Routers:* compute tables using routing algorithms, IP addresses, route aggregation
- *Switches:* Build their own forwarding table by recording source MAC addresses of all incoming frames



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A Brief Description of WiFi

Due to time constraints rely on notes and then book for clarification

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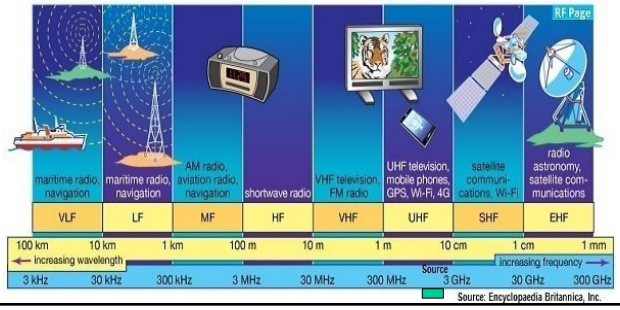
Electromagnetic Spectrum – wireless link!

2.4 GHz unlicensed band--shared by phones, WiFi, microwaves GPS, navigation systems, satellites, pagers, Bluetooth, television broadcasting, and most importantly GSM, CDMA and LTE mobile transmission.

>3 GHz more recently 5G higher https://www.rfpage.com/what-are-radio-frequency-bands-and-its-uses/



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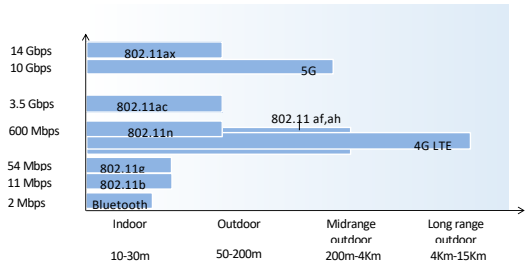
Characteristics of selected wireless links



WiFi is IEEE standard 802.11

Wireless and Mobile Networks: 7- 78

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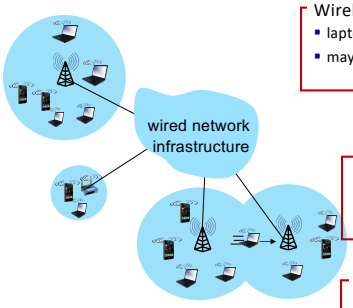
Elements of a wireless network



Wireless hosts

- laptop, smartphone, IoT
- may be stationary (non-mobile) or mobile

Base Station

- Relay – sends packets between wired network and wireless host(s) in its "area"

Wireless Link

- connects wireless devices to base station, also used as backbone link
- various transmission rates and distances, frequency bands for versions of WiFi (802.11 etc.)

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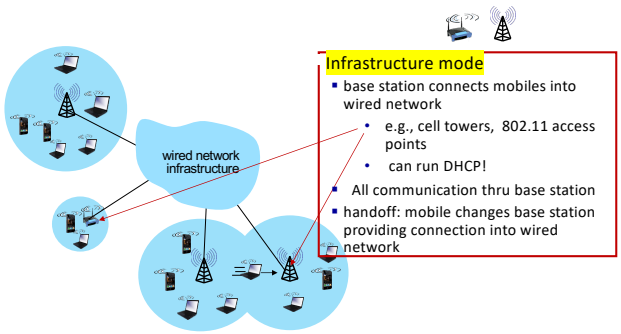
Elements of a wireless network



Infrastructure mode
- base station connects mobiles into wired network
  - e.g., cell towers, 802.11 access points
  - can run DHCP!
- All communication thru base station
- handoff: mobile changes base station providing connection into wired network

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802.11: Association with Base Station, Channels

- Spectrum divided into channels at different frequencies
  - Admin chooses frequency for AP
  - Interference possible: channel can be same as that chosen by neighboring AP!
- Arriving host: must associate with an AP
  - scans channels, listening for *beacon frames* containing AP's name (SSID) and MAC address
  - selects AP to associate with
  - then may perform authentication [Chapter 8]
  - then typically run DHCP to get IP address in AP's subnet

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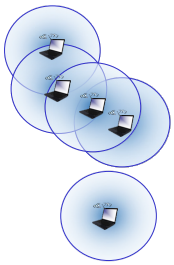
Elements of a wireless network



Ad hoc mode
- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves

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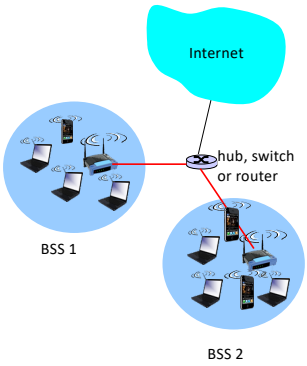
802.11 LAN architecture (WiFi)



Base Station (WiFi Access Point)
- Name (SSID)
- Access point can run DHCP!
- Wireless stations must connect to AP before data exchange
- All data goes through Access Point

Basic Service Set (BSS) (aka "cell")
- wireless hosts
- Access Point (AP)

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802.11 frame: addressing

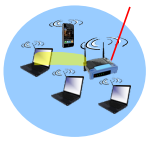
2	2	6	6	6	2	6	0 - 2312	4
frame control	duration	address 1	address 2	address 3	seq control	address 4	payload	CRC

Address 1: MAC address of wireless host or AP to receive this frame

Address 2: MAC address of wireless host or AP transmitting this frame

Address 3: MAC address of router interface to which AP is attached

Address 4: used only in ad hoc mode



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CSMA/CA – changes for WiFi

- Wireless channel challenges over shared link:
  - Decreased signal strength
  - Interference
  - Multipath propagation
  - High bit error rates
  - CSMA/CD doesn't work!
  - Cannot transmit AND listen simulataneously
- WiFi Protocol: CSMA/C(ollision)A(voidance)
  - IEEE 802.11
  - MAC protocol coordinates link access
    - Arbitrate access and avoid collisions
  - Uses CSMA - sense before transmitting, not during

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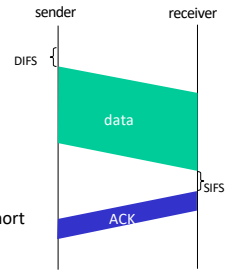
IEEE 802.11 MAC Protocol: CSMA/CA

802.11 sender

If channel sensed idle for DIFS (Distributed Interframe Space) then
- transmit entire frame (remember, no CD possible)
- if no ACK, retransmit

If sense channel busy then
- random backoff timer
- timer counts down while channel idle
- transmit when timer expires

802.11 receiver
- If frame received OK, return ACK after SIFS (Short Interframe Space)
  - ACK needed due to inability to sense collisions



- Why is ACK needed?

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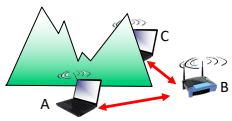
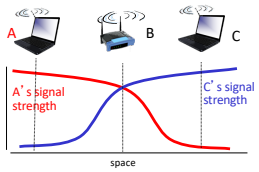
Big Wireless Network issues: Hidden Terminal and Signal Attenuation

Multiple wireless senders, receivers create additional problems (beyond multiple access):





Hidden terminal problem
- B is the Access Point
  - A,B,C are in the same subnet
  - B, A hear each other
  - B, C hear each other
- A, C can not hear each other
  - A, C unaware of their interference at B

Signal attenuation:
- B is the Access Point
  - A,B,C are in the same subnet
  - B, A hear each other
  - B, C hear each other
- A, C can not hear each other
  - A, C unaware of their interference at B

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Avoiding Collisions

Idea: sender "reserves" channel using small reservation packets

- Sender senses channel idle:
  - transmits *small* Request-to-Send (RTS) frame to Base Station (CSMA)
- If Base Station determines channel is free (CSMA):
  - broadcast Clear-to-Send (CTS) in response to RTS
- CTS heard by all nodes (why?)
  - CTS includes a reservation time
  - Sender transmits its data frame
  - Other stations defer transmissions during this time

Collisions avoided!

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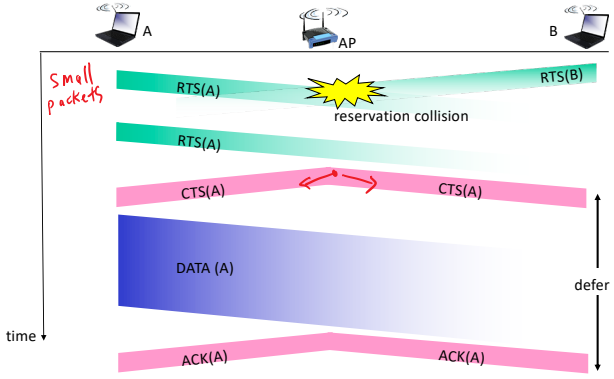
Collision Avoidance: RTS-CTS exchange



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